

SAFI, Morocco

WMO: 601850

Lat: 32.283N

Lon: 9.233W

Elev: 148

StdP: 14.62

Time Zone: 1.00 (E01)

Period: 90-12

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	42.6	44.5	36.6	32.0	55.2	39.0	35.1	54.0	29.8	59.4	23.5	62.5	7.3	60	0.443

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.9	91.0	68.8	86.3	68.7	83.3	68.2	72.1	82.4	71.2	80.0	70.4	78.4	12.6	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.6	109.5	74.3	68.8	106.2	73.9	68.0	103.4	73.5	35.9	81.9	35.1	80.5	34.4	78.2	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.7	19.3	17.7	DB	39.0	100.5	2.8	7.4	37.0	105.8	35.3	110.1	33.7	114.3	31.7	119.6
			WB	37.8	74.7	2.2	1.4	36.2	75.7	34.9	76.5	33.7	77.3	32.1	78.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.4	55.8	58.2	61.2	62.6	66.0	70.2	74.1	75.0	71.9	67.9	62.8	58.6	(6)
	DBStd	7.43	3.71	4.02	4.43	4.07	4.45	3.36	3.99	3.94	4.04	3.70	5.07	3.86	(7)
	HDD50	1	1	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	1059	284	193	135	94	34	0	0	0	1	13	103	202	(9)
	CDD50	5622	182	229	348	379	496	606	747	775	656	556	383	265	(10)
	CDD65	1208	0	3	18	23	65	156	282	310	208	104	36	3	(11)
	CDH74	6051	2	15	141	159	350	581	1517	1838	903	372	168	5	(12)
	CDH80	1666	0	0	29	29	109	138	485	569	205	61	41	0	(13)
(14) Wind	WSAvg	9.6	8.5	8.9	9.4	10.7	10.0	10.3	10.8	10.5	9.6	9.3	8.4	9.3	(14)
(15) Precipitation	PrecAvg	15.3	2.4	2.1	1.6	1.2	0.5	0.2	0.0	0.0	0.2	1.5	2.6	2.9	(15)
	PrecMax	40.1	18.0	9.5	5.4	4.3	2.5	0.9	0.1	0.1	0.8	5.4	9.1	11.1	(16)
	PrecMin	5.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
	PrecStd	6.3	3.2	1.8	1.3	1.0	0.7	0.2	0.0	0.0	0.2	1.6	2.5	2.6	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.8	76.2	83.0	83.9	90.7	91.0	101.2	96.9	90.4	85.2	84.8	73.6	(19)
		MCWB	59.9	61.2	61.7	63.7	65.7	69.5	70.7	69.8	67.3	65.9	64.0	58.4	(20)
	2%	DB	68.0	72.2	77.7	78.6	82.9	83.3	89.8	90.6	85.3	81.1	78.4	69.6	(21)
		MCWB	59.7	60.7	61.5	62.4	64.6	67.9	69.9	70.4	68.9	66.1	63.7	59.4	(22)
	5%	DB	65.7	69.4	74.0	74.1	77.7	79.8	84.6	85.3	81.7	77.9	73.7	67.4	(23)
		MCWB	58.4	59.6	60.6	61.4	63.9	67.1	69.7	70.0	67.9	65.1	63.1	60.0	(24)
	10%	DB	63.8	66.6	70.5	71.0	73.6	76.8	81.4	82.1	78.8	75.1	70.7	65.8	(25)
		MCWB	57.5	59.0	60.1	60.2	63.1	65.9	68.8	69.2	67.6	65.2	62.6	60.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.7	65.5	65.8	66.8	69.0	70.7	73.7	73.8	72.4	70.6	70.0	65.6	(27)
		MCDB	67.7	70.8	75.5	79.3	82.2	83.0	89.0	86.9	80.3	76.5	73.6	67.1	(28)
	2%	WB	62.3	62.8	63.9	64.2	66.7	69.2	71.8	72.3	71.2	69.1	67.8	63.9	(29)
		MCDB	66.0	67.8	72.2	72.6	76.5	78.8	82.8	83.0	78.5	74.1	71.6	66.2	(30)
	5%	WB	60.1	61.1	62.6	62.8	65.5	68.2	70.8	71.4	70.3	68.1	65.5	62.4	(31)
		MCDB	64.1	67.0	69.7	70.7	73.6	76.4	80.2	80.4	76.8	73.2	71.6	65.4	(32)
	10%	WB	58.3	59.6	61.3	61.7	64.4	67.3	69.9	70.6	69.5	67.0	63.8	60.9	(33)
		MCDB	62.4	65.5	68.3	68.9	71.4	74.7	78.4	78.8	75.5	72.2	69.9	64.6	(34)
(35) Mean Daily Temperature Range	MDBR		16.5	16.9	16.6	16.0	13.9	12.9	13.3	13.9	13.4	14.3	15.9	14.6	(35)
	5% DB	MCDBR	18.7	20.7	23.6	22.0	21.7	18.6	20.5	20.5	17.9	19.6	19.6	16.4	(36)
		MCWBR	11.1	11.0	11.9	9.7	7.8	6.3	5.8	5.7	6.1	8.0	9.1	9.3	(37)
	5% WB	MCDBR	14.1	17.7	19.1	17.5	16.0	15.0	16.3	15.6	13.1	14.0	14.2	10.5	(38)
		MCWBR	9.8	10.6	11.3	8.8	7.1	6.1	5.7	5.4	5.5	6.8	8.0	7.5	(39)
(40) Clear-Sky Solar Irradiance	taub		0.333	0.361	0.415	0.410	0.432	0.464	0.515	0.512	0.444	0.407	0.369	0.344	(40)
	taud		2.486	2.393	2.212	2.259	2.216	2.130	1.970	2.003	2.217	2.326	2.414	2.473	(41)
	Ebn at Noon		282	283	273	280	273	263	249	248	261	263	265	270	(42)
	Edh at Noon		28	34	43	43	46	49	58	55	43	36	30	27	(43)
(44) All-Sky Solar Radiation	RadAvg		1054	1336	1722	2078	2296	2419	2417	2198	1863	1403	1095	953	(44)
	RadStd		56	88	91	95	103	85	75	69	65	76	71	62	(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	+0.56	N/A	-1.57	+1.57	N/A	N/A	N/A	N/A	+207	+238

Nomenclature: See separate page